

Free-Field Computation of the $\mathbf{Vir} \oplus \mathbf{Vir}$ Branching Coefficient

Highest-vector prescription, norm, and three-point numerator

Machine working note

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Abstract

The human note reduces the all-NS superconformal block to two ordinary Virasoro blocks and leaves one quantity undetermined: the normalized three-point coefficient of the $\mathbf{Vir} \oplus \mathbf{Vir}$ highest-weight vectors v_n . We use the free-field highest-vector prescription suggested by arXiv:1111.2803, but every norm and three-point coefficient is evaluated with the BPZ, tensor-product, and fixed-parity conventions of the human note. Writing $k = 2n \in \mathbb{Z}$, the vector is a finite Fermi-sea product of the modes $\chi_r = f_r - i\vartheta_r$. The factorized ratio l in the paper is recorded only as a comparison; it is not identified with the human-note three-point function. The complete direct calculation at $n_i \in \{0, \pm\frac{1}{2}\}$ is given below. No Whittaker vector is used.

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1 NS free-field realization

We first fix the realization in which the branching vector will be computed. Set

$$Q = b + b^{-1}, \quad c = \frac{3}{2} + 3Q^2, \quad h(P) = \frac{1}{2} \left(\frac{Q^2}{4} - P^2 \right). \quad (1)$$

All modes in this section are NS modes: $m \in \mathbb{Z}$ and $r \in \mathbb{Z} + \frac{1}{2}$.

1.1 Oscillators and Fock vacuum

The auxiliary Majorana fermion has modes f_r ; this is the fermion denoted by ψ_r in the human note. The free-field realization of the SCA uses a Heisenberg oscillator c_m^ϵ , an NS fermion ϑ_r^ϵ , and a bosonic zero mode $\mathcal{P}$. Here $\epsilon \in \{+1, -1\}$ labels two *alternative reflected realizations* of the same SCA module; the $+$ and $-$ oscillators are not two simultaneous free fields. Their nonzero brackets are

$$\begin{aligned} [c_m^\epsilon, c_n^\epsilon] &= m\delta_{m+n,0}, & \{\vartheta_r^\epsilon, \vartheta_s^\epsilon\} &= \delta_{r+s,0}, & \{f_r, f_s\} &= \delta_{r+s,0}, \\ [\mathcal{P}, c_m^\epsilon] &= [\mathcal{P}, \vartheta_r^\epsilon] = 0, & [f_r, c_m^\epsilon] &= 0, & \{f_r, \vartheta_s^\epsilon\} &= 0. \end{aligned} \quad (2)$$

The auxiliary BPZ convention inherited from the human note is

$$f_r^\dagger = -f_{-r}. \quad (3)$$

Let $\mathbf{1}_F$ be annihilated by f_r for $r > 0$, and let $|P\rangle_\epsilon$ be annihilated by c_m^ϵ and ϑ_r^ϵ for $m, r > 0$, with $\mathcal{P}|P\rangle_\epsilon = P|P\rangle_\epsilon$. We write

$$|\text{vac}; P\rangle := \mathbf{1}_F \otimes |P\rangle_\epsilon. \quad (4)$$

Thus the tensor-product module is the span of arbitrary finite products of the negative modes f_{-r} , c_{-m}^ϵ , and ϑ_{-r}^ϵ acting on $|\text{vac}; P\rangle$.

1.2 SCA generators

In the realization labelled by ϵ , the abstract SCA generators act for $m \neq 0$ as

$$\begin{aligned} L_m &= \frac{1}{2} \sum_{j \neq 0, m} c_j^\epsilon c_{m-j}^\epsilon + \frac{1}{2} \sum_{r \in \mathbb{Z} + \frac{1}{2}} \left(r - \frac{m}{2} \right) \vartheta_{m-r}^\epsilon \vartheta_r^\epsilon + \frac{i}{2} (Qm + 2\epsilon \mathcal{P}) c_m^\epsilon, \\ G_r &= \sum_{m \neq 0} c_m^\epsilon \vartheta_{r-m}^\epsilon + i(Qr + \epsilon \mathcal{P}) \vartheta_r^\epsilon, \\ L_0 &= \sum_{m > 0} c_{-m}^\epsilon c_m^\epsilon + \sum_{r > 0} r \vartheta_{-r}^\epsilon \vartheta_r^\epsilon + \frac{1}{2} \left(\frac{Q^2}{4} - \mathcal{P}^2 \right). \end{aligned} \quad (5)$$

These formulas obey the $\mathcal{N} = 1$ SCA relations at the central charge in Eq. (1), and $L_0|P\rangle_\epsilon = h(P)|P\rangle_\epsilon$. The realization $\epsilon = -1$ is used for $k > 0$, and $\epsilon = +1$ for $k < 0$. They are related by the super-Liouville reflection operator.

It is now meaningful to define the isotropic fermionic modes

$$\chi_r^\epsilon := f_r - i\vartheta_r^\epsilon, \quad \{\chi_r^\epsilon, \chi_s^\epsilon\} = 0. \quad (6)$$

The vanishing anticommutator is the reason that the highest vector below is a finite Fermi-sea product.

1.3 The commuting Virasoro pair

For completeness, define the auxiliary-fermion stress tensor and the mixed bilinear

$$L_n^F = \frac{1}{2} \sum_{r \in \mathbb{Z} + \frac{1}{2}} r : f_{n-r} f_r :, \quad U_n = \sum_{r \in \mathbb{Z} + \frac{1}{2}} f_{n-r} G_r. \quad (7)$$

The two commuting Virasoro generators used in the human note are

$$\begin{aligned} L_n^{(1)} &= \frac{b^{-1}}{b^{-1} - b} L_n - \frac{b^{-1} + 2b}{b^{-1} - b} L_n^F + \frac{1}{b^{-1} - b} U_n, \\ L_n^{(2)} &= \frac{b}{b - b^{-1}} L_n - \frac{b + 2b^{-1}}{b - b^{-1}} L_n^F + \frac{1}{b - b^{-1}} U_n. \end{aligned} \quad (8)$$

They obey

$$[L_m^{(1)}, L_n^{(2)}] = 0, \quad c^{(i)} = 1 + 6(Q^{(i)})^2, \quad Q^{(i)} = b^{(i)} + (b^{(i)})^{-1}, \quad (9)$$

where

$$b^{(1)} = \frac{2b}{\sqrt{2 - 2b^2}}, \quad (b^{(2)})^{-1} = \frac{2b^{-1}}{\sqrt{2 - 2b^{-2}}}. \quad (10)$$

This finishes the algebraic setup. Every branching vector computed below is an explicit vector in the Fock space of Eqs. (2)–(4).

2 The quantity left open in the human note

The human note uses

$$\mathcal{M}_{\text{NS}}(h) = \mathcal{V}_{F_{1/2}} \otimes \mathcal{V}_{\text{SCA}_{1/2}}(h), \quad h = \frac{1}{2} \left(\frac{Q^2}{4} - P^2 \right), \quad Q = b + b^{-1}, \quad (11)$$

with the auxiliary free-fermion factor first. The branching rule is

$$\mathcal{M}_{\text{NS}}(h) \cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} \mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)}), \quad h_n^{(1)} + h_n^{(2)} = h + 2n^2. \quad (12)$$

The corresponding highest-weight vector is denoted by v_n , and its parity is $2n \pmod{2}$.

For the ordered trinion used throughout the human note, slot 1 is the BPZ-conjugate state at infinity, slot 2 is inserted at 1, and slot 3 is the ket at 0. In precisely this order the human note defines

$$B_a(h_1, h_2, h_3; n_1, n_2, n_3) = \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})}{\|v_{1,n_1}\| \|v_{2,n_2}\| \|v_{3,n_3}\|}, \quad a = 2(n_1 + n_2 + n_3) \pmod{2}. \quad (13)$$

Only B_a^2 enters the conformal-block decomposition. Directly from the human definition,

$$[B_a(h_1, h_2, h_3; n_1, n_2, n_3)]^2 = \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})^2}{N_{k_1}(P_1)N_{k_2}(P_2)N_{k_3}(P_3)}, \quad k_i = 2n_i, \quad N_{k_i}(P_i) = \|v_{i,n_i}\|^2. \quad (14)$$

No reversal of the labels $1 \leftrightarrow 3$ is made in this note. Other slot orders must be obtained with the Möbius and Koszul conventions already fixed in the human note.

3 Highest vector from the free-field realization

3.1 The Fermi-sea vector

Set $k = 2n \in \mathbb{Z}$. For $k \neq 0$, the prescription is

$$|P, k\rangle = \Omega_k(P) \prod_{j=1}^{|k|} \chi_{-(2j-1)/2}^{-\text{sgn}(k)} |\text{vac}; P\rangle, \quad |P, 0\rangle = |\text{vac}; P\rangle. \quad (15)$$

The product is ordered with increasing j from left to right. Its total level is

$$\sum_{j=1}^{|k|} \frac{2j-1}{2} = \frac{k^2}{2} = 2n^2, \quad (16)$$

which is precisely the shift in Eq. (12).

The factor $\Omega_k(P)$ is fixed by the standard SCA-PBW normalization

$$|P, k\rangle = \left(G_{-1/2}^{k^2} + \text{terms of lower degree in } G \right) |P\rangle. \quad (17)$$

This fixes the vector, including its scale, without referring to a Whittaker state.

The commutators quoted in the literature are

$$\begin{aligned} [L_m^{(1)} + L_m^{(2)}, \chi_r^\epsilon] &= -\left(\frac{m}{2} + r\right) \chi_{m+r}^\epsilon, \\ [bL_m^{(1)} + b^{-1}L_m^{(2)}, \chi_r^\epsilon] &= -((m+r)Q + \epsilon\mathcal{P}) \chi_{m+r}^\epsilon + i \sum_{j \neq 0} c_j^\epsilon \chi_{m+r-j}^\epsilon. \end{aligned} \quad (18)$$

They prove directly that Eq. (15) is killed by both positive Virasoro algebras. The two weights are

$$\begin{aligned} h_n^{(1)} &= \frac{1}{4}(Q^{(1)})^2 - \left(\frac{P}{\sqrt{2-2b^2}} + nb^{(1)} \right)^2, \\ h_n^{(2)} &= \frac{1}{4}(Q^{(2)})^2 - \left(\frac{P}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1} \right)^2. \end{aligned} \quad (19)$$

The first nontrivial vectors, already converted back to the human note's SCA and auxiliary-fermion modes, are

$$\begin{aligned} |P, 1\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} + P \right) f_{-1/2} \right] |P, 0\rangle, \\ |P, -1\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} - P \right) f_{-1/2} \right] |P, 0\rangle. \end{aligned} \quad (20)$$

They are the human-note vectors $v_{1/2}$ and $v_{-1/2}$.

4 Factorized three-point numerator

4.1 Dictionary for an ordered trinion

Parameterize the inserted NS weight by

$$h_2 = \frac{1}{2}\alpha_2(Q - \alpha_2), \quad \alpha_2 = \frac{Q}{2} + P_2. \quad (21)$$

For the ordered matrix element

$$\langle P_1, k_1 | \mathbb{V}_{\alpha_2}^{(k_2)}(1) | P_3, k_3 \rangle, \quad (22)$$

the state-field correspondence identifies $\mathbb{V}_{\alpha_2}^{(k_2)}$ with the same highest vector $|P_2, k_2\rangle$ used on the middle leg. The normalized ratio in arXiv:1111.2803 is

$$l(\alpha, m | P', k', P, k) = \begin{cases} \frac{\langle P', k' | \mathbb{V}_{\alpha}^{(m)} | P, k \rangle}{\langle P' | \mathbb{V}_{\alpha}^{(0)} | P \rangle}, & m + k + k' \text{ even}, \\ \frac{\langle P', k' | \mathbb{V}_{\alpha}^{(m)} | P, k \rangle}{\langle P' | \mathbb{V}_{\alpha}^{(\pm 1)} | P \rangle}, & m + k + k' \text{ odd}. \end{cases} \quad (23)$$

The corresponding primary denominators have the same numerical normalization as the human note: $\rho_0(\phi_1, \phi_2, \phi_3) = 1$ in the even case and $\rho_1(\phi_1, G_{-1/2}\phi_2, \phi_3) = 1$ in the odd case. Define the literature ratio in the human slot labels by

$$L_a^{\text{lit}}(1, 2, 3) := l(\alpha_2, k_2 | P_1, k_1, P_3, k_3), \quad a = k_1 + k_2 + k_3 \pmod{2}. \quad (24)$$

Matching the two primary denominators does *not* by itself prove

$$T_a^{\text{H}} = L_a^{\text{lit}}(1, 2, 3). \quad (25)$$

Equation (25) is a convention-sensitive statement about the complete highest vectors. It must also match the auxiliary fermion BPZ sign, the fixed-parity conversion in the human note, and the state-field map for the middle vector. This is the central audit below.

4.2 The product formula

For $\sigma, \tau \in \{+1, -1\}$, define

$$x_{\sigma\tau} := \alpha + \sigma P' + \tau P, \quad r_{\sigma\tau} := \frac{m + \sigma k' + \tau k}{2}. \quad (26)$$

For a nonnegative integer r , set

$$\begin{aligned} s_{\text{even}}(x, r) &= 2^{-r^2/2} \prod_{\substack{i,j \geq 1, \ i+j \leq 2r \\ i+j \equiv 0 \pmod{2}}} \left[x + (i-1)b + (j-1)b^{-1} \right], \\ s_{\text{odd}}(x, r) &= 2^{-r(r+1)/2} \prod_{\substack{i,j \geq 1, \ i+j \leq 2r+1 \\ i+j \equiv 1 \pmod{2}}} \left[x + (i-1)b + (j-1)b^{-1} \right]. \end{aligned} \quad (27)$$

For negative integer r , extend them by

$$s_{\text{even}}(x, r) = (-1)^r s_{\text{even}}(Q - x, -r), \quad s_{\text{odd}}(x, r) = s_{\text{odd}}(Q - x, -r). \quad (28)$$

Finally define

$$\text{Int}(x) := \text{sgn}(x) \lfloor |x| \rfloor. \quad (29)$$

The literature ratio is then

$$l(\alpha, m \mid P', k', P, k) = \begin{cases} \prod_{\sigma, \tau = \pm 1} s_{\text{even}}(x_{\sigma\tau}, r_{\sigma\tau}), & m + k + k' \text{ even}, \\ \prod_{\sigma, \tau = \pm 1} s_{\text{odd}}(x_{\sigma\tau}, \text{Int}(r_{\sigma\tau})), & m + k + k' \text{ odd}. \end{cases} \quad (30)$$

This is the factorized matrix-element ratio of arXiv:1111.2803. It becomes the desired human-note numerator only after Eq. (25), including any convention factor, has been established.

4.3 Endpoint order and the first convention check

The fixed-parity trilinear form used in the human note obeys the endpoint reflection rule

$$\widehat{\rho}_a(x_1, x_2, x_3) = (-1)^{|x_2|(|x_1|+|x_3|)} \widehat{\rho}_a(x_3, x_2, x_1), \quad (31)$$

with the weights and labels in slots (1) and (3) exchanged on the right-hand side. For the branching primaries this sign is $(-1)^{k_2(k_1+k_3)}$. Thus the reversed expression $\widehat{\rho}_a(v_{3,n_3}, v_{2,n_2}, v_{1,n_1})$, used in an earlier version of this note, is not the definition in the human note and cannot be substituted without this reflection sign.

Already at $n_i = 0, \pm \frac{1}{2}$, one must expand the explicit vectors

$$v_{i, \epsilon_i/2} = G_{-1/2} \phi_i + a_i f_{-1/2} \phi_i, \quad a_i := \frac{Q}{2} + \epsilon_i P_i, \quad \epsilon_i = \pm 1, \quad (32)$$

inside the tensor-product trilinear form of the human note. Its auxiliary BPZ convention gives

$$\rho^F(f\mathbf{1}, f\mathbf{1}, \mathbf{1}) = -1, \quad \rho^F(f\mathbf{1}, \mathbf{1}, f\mathbf{1}) = -1, \quad \rho^F(\mathbf{1}, f\mathbf{1}, f\mathbf{1}) = +1. \quad (33)$$

Using the human note's fixed-parity component table then gives

$$\begin{aligned} T_0^{\text{H}}[\epsilon_1/2, \epsilon_2/2, 0] &= -\frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right], \\ T_0^{\text{H}}[\epsilon_1/2, 0, \epsilon_3/2] &= \frac{1}{2} \left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3 \right)^2 \right], \\ T_0^{\text{H}}[0, \epsilon_2/2, \epsilon_3/2] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right]. \end{aligned} \quad (34)$$

The product formula for L^{lit} has the same last two entries but the opposite sign in the first entry. Equivalently, on these cases,

$$T_a^{\text{H}} = (-1)^{k_1 k_2} L_a^{\text{lit}}. \quad (35)$$

This establishes a genuine convention factor at the first nontrivial level. Whether Eq. (35) is the complete all-level translation must be checked with an explicit $|k| = 2$ vector; it is not assumed below for an unsquared coefficient.

5 Norm from the same formula

Take the inserted field to be the identity:

$$\alpha = 0, \quad m = 0, \quad P' = P, \quad k' = k. \quad (36)$$

Then Eq. (23) becomes the norm, while the four factors in Eq. (30) reduce to

$$(x_{++}, r_{++}) = (2P, k), \quad (x_{+-}, r_{+-}) = (0, 0), \quad (x_{-+}, r_{-+}) = (0, 0), \quad (x_{--}, r_{--}) = (-2P, -k). \quad (37)$$

Since $s_{\text{even}}(x, 0) = 1$, we obtain

$$\boxed{N_k(P) = \langle P, k | P, k \rangle = s_{\text{even}}(2P, k) s_{\text{even}}(-2P, -k)}. \quad (38)$$

Equivalently, for $p \geq 0$,

$$\begin{aligned} N_p(P) &= (-1)^p s_{\text{even}}(2P, p) s_{\text{even}}(Q + 2P, p), \\ N_{-p}(P) &= (-1)^p s_{\text{even}}(Q - 2P, p) s_{\text{even}}(-2P, p). \end{aligned} \quad (39)$$

At level 1/2, Eq. (20) gives an independent direct check. Using

$$\langle G_{-1/2} \phi, G_{-1/2} \phi \rangle = 2h, \quad \langle f_{-1/2} \mathbf{1}, f_{-1/2} \mathbf{1} \rangle = -1, \quad (40)$$

and vanishing mixed inner product, one finds

$$\begin{aligned} N_1(P) &= 2h - \left(\frac{Q}{2} + P \right)^2 = -P(Q + 2P), \\ N_{-1}(P) &= 2h - \left(\frac{Q}{2} - P \right)^2 = P(Q - 2P). \end{aligned} \quad (41)$$

These agree with Eq. (38), because $s_{\text{even}}(x, 1) = x/\sqrt{2}$. The same computation also gives

$$\langle P, 1 | P, -1 \rangle = 0. \quad (42)$$

6 Squared coefficient obtained from the literature ratio

The product formula and the norm determine the following literature-normalized candidate:

$$\mathcal{B}_a^{\text{lit}}(P_1, P_2, P_3; k_1, k_2, k_3) = \frac{l\left(\frac{Q}{2} + P_2, k_2 \mid P_1, k_1, P_3, k_3\right)^2}{N_{k_1}(P_1)N_{k_2}(P_2)N_{k_3}(P_3)}, \quad (43)$$

where l is Eq. (30), N is Eq. (38), and

$$k_i = 2n_i, \quad a = k_1 + k_2 + k_3 \pmod{2}. \quad (44)$$

At the cases checked in Eq. (34), the translation is a sign and hence $\mathcal{B}_a^{\text{lit}} = \mathcal{B}_a$. The all-level identification requires proving that the translation between T_a^{H} and L_a^{lit} remains a pure sign for the explicit higher branching vectors.

If an unsquared B_a is desired, one must choose square roots of the three complex Shapovalov norms, fix an overall phase for every branch vector, and include the human-to-literature trilinear convention factor. Principal square roots alone do not define the human-note coefficient.

7 Low-level values

The product formula immediately gives

$$l(\alpha, 0 \mid P', 0, P, 0) = 1, \quad l(\alpha, 1 \mid P', 0, P, 0) = 1. \quad (45)$$

The first is the even primary normalization; the second is the odd normalization.

With one $k = 1$ branch vector and the other two labels zero,

$$l(\alpha, 0 \mid P', 0, P, 1) = l(\alpha, 0 \mid P', 1, P, 0) = 1. \quad (46)$$

Consequently the corresponding squared normalized coefficient is simply the inverse norm of that leg.

With two $k = 1$ vectors, the even numerators are

$$\begin{aligned} l(\alpha, 0 \mid P', 1, P, 1) &= \frac{1}{2}(\alpha + P' + P)(\alpha - P' - P - Q), \\ l(\alpha, 1 \mid P', 1, P, 0) &= \frac{1}{2}(\alpha + P' + P)(\alpha + P' - P), \\ l(\alpha, 1 \mid P', 0, P, 1) &= \frac{1}{2}(\alpha + P' + P)(\alpha - P' + P). \end{aligned} \quad (47)$$

These are the first nontrivial polynomial checks to reproduce directly from the human-note Ward identities.

For example,

$$\mathcal{B}_0^{\text{lit}}(P_1, P_2, P_3; 1, 0, 1) = \frac{(\alpha_2 + P_3 + P_1)^2(\alpha_2 - P_3 - P_1 - Q)^2}{4N_1(P_3)N_1(P_1)}, \quad \alpha_2 = \frac{Q}{2} + P_2. \quad (48)$$

7.1 Complete unpacking for $n_i = 0, \pm \frac{1}{2}$

We now remove the product notation completely at the first nontrivial branching level. In this subsection the slot order (n_1, n_2, n_3) is exactly the human order: BPZ-conjugate state at infinity, inserted field at one, and ket at zero. To keep the left-hand sides readable, define

$$\mathcal{B}_a^{\text{lit}}[n_1, n_2, n_3] := \mathcal{B}_a^{\text{lit}}(P_1, P_2, P_3; 2n_1, 2n_2, 2n_3), \quad L_a^{\text{lit}}[n_1, n_2, n_3] := l\left(\frac{Q}{2} + P_2, 2n_2 \mid P_1, 2n_1, P_3, 2n_3\right). \quad (49)$$

The human numerator remains $T_a^{\text{H}}[n_1, n_2, n_3] = \hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})$.

Whenever $n_i \neq 0$, write

$$\epsilon_i := 2n_i \in \{+1, -1\}. \quad (50)$$

The free-field vector and its norm take the uniform form

$$\begin{aligned} |P, \epsilon\rangle &= \left[G_{-1/2} + \left(\frac{Q}{2} + \epsilon P \right) f_{-1/2} \right] |P, 0\rangle, \\ \langle P, \epsilon | P, \epsilon \rangle &= -\epsilon P (Q + 2\epsilon P). \end{aligned} \quad (51)$$

Zero or one nonzero label. For the vacuum branch,

$$T_0^{\text{H}}[0, 0, 0] = L_0^{\text{lit}}[0, 0, 0] = 1, \quad \mathcal{B}_0^{\text{lit}}[0, 0, 0] = 1. \quad (52)$$

If exactly one label is nonzero, the numerator remains one and

$$\begin{aligned} T_1^{\text{H}}[\epsilon_1/2, 0, 0] &= T_1^{\text{H}}[0, \epsilon_2/2, 0] = T_1^{\text{H}}[0, 0, \epsilon_3/2] \\ &= L_1^{\text{lit}}[\epsilon_1/2, 0, 0] = L_1^{\text{lit}}[0, \epsilon_2/2, 0] = L_1^{\text{lit}}[0, 0, \epsilon_3/2] = 1, \\ \mathcal{B}_1^{\text{lit}}[\epsilon_1/2, 0, 0] &= -\frac{\epsilon_1}{P_1(Q + 2\epsilon_1 P_1)}, \\ \mathcal{B}_1^{\text{lit}}[0, \epsilon_2/2, 0] &= -\frac{\epsilon_2}{P_2(Q + 2\epsilon_2 P_2)}, \\ \mathcal{B}_1^{\text{lit}}[0, 0, \epsilon_3/2] &= -\frac{\epsilon_3}{P_3(Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (53)$$

For example, a $+\frac{1}{2}$ label gives $-[P_i(Q + 2P_i)]^{-1}$, whereas a $-\frac{1}{2}$ label gives $[P_i(Q - 2P_i)]^{-1}$.

Exactly two nonzero labels. When the bra label vanishes, the literature ratio is

$$\begin{aligned} L_0^{\text{lit}}[0, \epsilon_2/2, \epsilon_3/2] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right], \\ \mathcal{B}_0^{\text{lit}}[0, \epsilon_2/2, \epsilon_3/2] &= \frac{\left[\left(\frac{Q}{2} + \epsilon_2 P_2 + \epsilon_3 P_3 \right)^2 - P_1^2 \right]^2}{4\epsilon_2 \epsilon_3 P_2 P_3 (Q + 2\epsilon_2 P_2) \times (Q + 2\epsilon_3 P_3)}. \end{aligned} \quad (54)$$

Here $T_0^{\text{H}} = L_0^{\text{lit}}$, as shown directly in Eq. (34). When the inserted label vanishes, the literature ratio is

$$L_0^{\text{lit}}[\epsilon_1/2, 0, \epsilon_3/2] = \frac{1}{2} \left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3 \right)^2 \right],$$

$$\mathcal{B}_0^{\text{lit}}[\epsilon_1/2, 0, \epsilon_3/2] = \frac{\left[P_2^2 - \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_3 P_3\right)^2\right]^2}{4\epsilon_1 \epsilon_3 P_1 P_3 (Q + 2\epsilon_1 P_1) \times (Q + 2\epsilon_3 P_3)}. \quad (55)$$

Again $T_0^{\text{H}} = L_0^{\text{lit}}$. Finally, when the ket label vanishes, the literature ratio is

$$\begin{aligned} L_0^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, 0] &= \frac{1}{2} \left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right], \\ \mathcal{B}_0^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, 0] &= \frac{\left[\left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 \right)^2 - P_3^2 \right]^2}{4\epsilon_1 \epsilon_2 P_1 P_2 (Q + 2\epsilon_1 P_1) \times (Q + 2\epsilon_2 P_2)}. \end{aligned} \quad (56)$$

In this last case the human numerator has the opposite sign:

$$T_0^{\text{H}}[\epsilon_1/2, \epsilon_2/2, 0] = -L_0^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, 0]. \quad (57)$$

The sign is fixed by the human BPZ and fixed-parity conventions; it is not a freely chosen phase. It disappears after squaring this numerator.

Three nonzero labels. All eight sign choices are covered by

$$\begin{aligned} L_1^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, \epsilon_3/2] &= \frac{1}{2} \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b \right) \\ &\quad \times \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b^{-1} \right), \end{aligned} \quad (58)$$

$$\begin{aligned} \mathcal{B}_1^{\text{lit}}[\epsilon_1/2, \epsilon_2/2, \epsilon_3/2] &= -\frac{\epsilon_1 \epsilon_2 \epsilon_3 \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b \right)^2}{4P_1 P_2 P_3 (Q + 2\epsilon_1 P_1)(Q + 2\epsilon_2 P_2) \times (Q + 2\epsilon_3 P_3)} \\ &\quad \times \left(\frac{Q}{2} + \epsilon_1 P_1 + \epsilon_2 P_2 + \epsilon_3 P_3 + b^{-1} \right)^2. \end{aligned} \quad (59)$$

The displayed L_1^{lit} is not yet identified with T_1^{H} . This is the first case in the 27-entry table where the convention translation must be checked using all terms in the tensor-product highest vectors. Equations (52)–(59) exhaust all 27 literature expressions $n_i \in \{0, \pm \frac{1}{2}\}$. They are the formulas evaluated by `Code/two_virasoro_fusion.py`; its unsquared output should therefore be read as L_a^{lit} , not yet as T_a^{H} .

8 Ramond extension: status of the literature

This section records the literature status as of 20 August 2026. It is important to distinguish three increasingly strong statements:

1. the decomposition of an $F_0 \oplus \text{SCA}_0$ Ramond module into two-Virasoro modules;
2. a norm or Whittaker overlap of one two-Virasoro highest vector; and

3. the generic, chiral (NS, R, R) three-point coefficient, including its Ramond-ground-state phases.

Only the third statement supplies the coefficient required by the human note. In its notation the target is

$$\mathbb{B}_f^{(\eta)}(h_1, \beta_2, \beta_3; n_1, n_2, n_3) = \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3})}{\|v_{1,n_1}\| \|v_{2,n_2}^{\alpha_2}\| \|v_{3,n_3}^{\alpha_3}\|}, \quad (60)$$

where $n_1 \in \frac{1}{2}\mathbb{Z}$, $n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$, and the parity labels and the chiral sign η must be retained.

8.1 What has actually been established

Gauge-theory and block evidence. Ito’s ALE-space proposal, arXiv:1110.2176, identifies Ramond Whittaker norms and overlaps with instanton partition functions. Belavin–Mukhametzhanov, arXiv:1210.7454, similarly obtain four-point blocks containing Ramond fields from AGT data. These works provide strong independent control of Ramond blocks, but neither computes Eq. (60): their algebra and basis are not the present $\text{Vir} \oplus \text{Vir} \subset \text{F} \oplus \text{SCA}$ branching basis, and they do not determine its chiral three-point normalization.

The all-label double-Liouville candidate. Schomerus–Suchanek, arXiv:1210.1856, construct an infinite tower $\Phi_\alpha^{(k)}$ in super-Liouville theory times a free fermion. Integer k belongs to the NS sector and half-integer k to the Ramond sector. They propose that every $\Phi_\alpha^{(k)}$ is proportional to a primary of the two bosonic Liouville theories; this is precisely the local-field version of the branching-primary identification needed here. They give a candidate normalization and test several nontrivial local three-point functions, including Ramond labels $k = \pm\frac{1}{2}$ and $\pm\frac{3}{2}$.

This is the closest result in the literature to a formula for the full Ramond branching coefficient, but it is not yet Eq. (60). The paper explicitly does not prove the three-point identity for arbitrary labels. Its checks concern local, nonchiral correlators and leave a fourth root of unity $\varpi(k_3, k_2, k_1)$, with $\varpi^4 = 1$. Consequently it does not fix the separate holomorphic coefficient, the two Ramond parity copies, or the phases associated with a convention such as

$$G_0 w_\beta^\pm = i\beta e^{\mp i\pi/4} w_\beta^\mp. \quad (61)$$

It should be used below as an all-label candidate and a magnitude check, not as an already proved chiral branching formula.

The screening-charge proof stops in the NS sector. Hadasz–Jaskólski, arXiv:1312.4520, formulate the correspondence in terms of chiral structure constants and give a complete proof in the NS sector. Their reflection map, screening-charge matrix elements, singular-vector residues, and factorized chiral constants are the direct template for the calculation proposed below. They cite the Ramond tests of Schomerus–Suchanek and remark that a Ramond extension appears straightforward, but they do not carry out that extension. In particular, they do not introduce the Ramond zero-mode Clifford module needed to resolve the two parity copies in Eq. (60).

The Ramond module and highest vectors are known. Bershtein–Shchepochkin, arXiv:1608.02568, prove the generic Ramond branching rule

$$\mathcal{M}_R(\beta) \cong \bigoplus_{\epsilon=0,1} \bigoplus_{n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}} \mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)}), \quad (62)$$

and give explicit highest vectors as finite products of integer modes $\chi_r^\pm = f_r - i\vartheta_r^\pm$, including the zero-mode factors that distinguish the two parity copies. Thus the vectorial input for a direct free-field calculation is available.

The same paper does *not* obtain generic three-point matrix elements. For the much weaker irregular/Whittaker overlap it states a conjectural squared coefficient in terms of the odd product s_{odd} . The conjecture was checked by computer for $|n| \leq 9/4$ (and in the reported series test through relative order ten), while its sign remains undetermined. This supplies a valuable norm/irregular limit check, but not the coefficient in Eq. (60).

The recent screening algebra does not close the gap. Ohkubo, arXiv:2505.02926, constructs a q -deformation of $F \oplus \text{SCA}$, two commuting q -Virasoro algebras, and screening currents whose degeneration gives the super-Virasoro screenings. The representation theory in that paper is NS. Its conclusion says that a modified Fock module and half-integer highest-weight charge should allow the Ramond sector; no Ramond branching coefficient is computed.

8.2 Literature verdict and the remaining problem

No source found in the literature through the date above derives the generic chiral coefficient $\mathbb{B}_f^{(n)}$ for all branching labels while also fixing the two Ramond ground-state copies and their phases. The present status is therefore

Ramond branching rule and explicit highest vectors: proved, all-label local double-Liouville formula: proposed and tested, generic chiral branching coefficient with phases: open.	(63)
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This conclusion is narrower than saying that Ramond blocks or Ramond super-Liouville three-point functions are unknown: both are known. What is missing is their convention-complete projection onto the specific two-Virasoro branching highest vectors.

8.3 Screening-charge route to be pursued here

The most economical derivation is now well localized:

1. Use the integer-mode products of arXiv:1608.02568 as the definitions of the two Ramond branching vectors $v_n^\epsilon(P)$, including their $\chi_0^\pm$ factors.
2. Extend the reflected-Fock-space BPZ pairing used in the NS calculation to the four-dimensional zero-mode ground space $F_0 \otimes \text{SCA}_0$. The pairing must implement Eq. (61), rather than identify the two parity copies implicitly.

3. Generalize the screening-charge matrix elements of arXiv:1312.4520 from NS exponentials to the spin-field intertwiners of an (NS, R, R) vertex. Charge neutrality fixes the number of screenings; the zero-mode Clifford matrix fixes which of the two chiral Ramond structures is selected.
4. Evaluate the resulting finite Selberg-type products and divide by the three reflected-Fock norms. The product must reduce to the Schomerus–Suchanek double-Liouville candidate after multiplying the two chiral halves and forgetting the zero-mode phase.
5. Anchor the remaining square-root choices at $n_2, n_3 = \pm \frac{1}{4}$ by a direct ground-state Ward-identity calculation, then propagate them by the screening shifts. In the Whittaker limit the square must reproduce the s_{odd} formula conjectured and tested in arXiv:1608.02568.

The first new calculation should therefore be the four-by-four matrix of ground-level (NS, R, R) branching numerators. It will separate a genuine dynamical product formula from the convention-only zero-mode phases before higher integer χ_{-r} modes are introduced.

9 What remains to be audited

The calculation above settles the all-NS coefficient, modulo a direct audit of the literature product formula against the human note’s ordered Ward identities. The efficient audit is:

1. reproduce Eqs. (41) from the Gram form;
2. reproduce Eqs. (46) and Eq. (47) from the three-point Ward identities;
3. check a $|k| = 2$ case using the explicit level-two vectors in Appendix B of arXiv:1111.2803; and
4. only then use Eq. (43) as the all-level input to the block decomposition.

The Ramond branching vectors have an analogous free-field prescription in arXiv:1608.02568. It uses integer $\chi_r^\pm$ modes and has two parity copies. Their norms and (NS, R, R) three-point coefficients require a separate translation to the human note’s phase convention

$$G_0 w_\beta^\pm = i\beta e^{\mp i\pi/4} w_\beta^\mp. \quad (64)$$

No Whittaker coefficient is used here.

10 Sources

The free-field vector, its PBW normalization, the blow-up factors, and the functions $s_{\text{even}}, s_{\text{odd}}$ are in arXiv:1111.2803, especially Sec. 3.2, Proposition 3.2, and Appendix B. The later presentation and the Ramond analogue are in arXiv:1608.02568, Sec. 4.

The Ramond local-field candidate and its checked three-point examples are in arXiv:1210.1856, Secs. 4.2–4.3. The chiral screening/reflection proof that presently exists only in the NS sector is arXiv:1312.4520. Indirect Ramond Whittaker and block evidence comes from arXiv:1110.2176 and arXiv:1210.7454. The recent q -deformed algebra and screening-current construction, which leaves its Ramond representation theory for future work, is arXiv:2505.02926.